AdaM-Rec: Adaptive Modality Routing for Multimodal Recommendation

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Abstract

While recent multimodal recommender systems have demonstrated the effectiveness of incorporating visual and textual information to improve downstream performance, most existing methods rely on static modality fusion, assuming that the relative importance of textual and visual signals remains stable across recommendation scenarios. This design may not fully account for an important variation across recommendation requests: some queries require fine-grained visual cues, whereas others are better served by textual or functional semantics, in which case indiscriminate modality fusion brings in uninformative cues and impairs recommendation quality. To address this, we propose **AdaM-Rec**, an LLM-based framework for adaptive modality routing in multimodal recommendation, which enables dynamic calibration of reliance on textual and multimodal evidence for user-specific queries. Built on structured natural-language representations of items and user preferences, it estimates modality reliability using proxy recall tasks. Specifically, it generates pseudo-queries that match the granularity of the actual query while pointing to the user's positively interacted items as verifiable proxy targets, evaluating which modality yields better recall performance in analogous scenarios and optimizing the routing strategy in an agentic manner. It then performs routed recall with optimized strategy, enriches results with collaborative items, and ranks candidates by their relevance to both the query and user preferences. Experiments demonstrate that AdaM-Rec delivers strong performance against state-of-the-art baselines, highlighting the effectiveness and broader potential of adaptive control over modality reliance in multimodal recommendation.

1 Introduction

In e-commerce, user decisions are naturally shaped by multimodal product information, ranging from textual attributes such as descriptions and specifications to visual characteristics related to aesthetics and appearance [47, 64]. This motivates increasing research efforts in multimodal recommender systems, which have enhanced item profiling and user modeling by incorporating textual and visual information through representation learning [39, 68], multimodal tokenization [67], and multimodal large language models (MLLMs) [27, 30]. However, the effectiveness of visual information is not universal [76]; rather, it depends on the extent to which a product's visual attributes align with the user's intent. As illustrated in Figure 1, the visual information of products could be essential for appearance-driven queries such as "sneakers with a thick wavy sole," but may provide little evidence for functionality-driven queries like "70W MacBook USB-C charger." In such cases, product images may even introduce noise by retrieving visually similar yet functionally mismatched items [74, 76].

This query-level variability in modality reliance poses a fundamental challenge for existing multimodal recommenders. Although prior studies have shown the benefits of incorporating both textual

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Codes are available at https://github.com/RomGai/AdaM-Rec.

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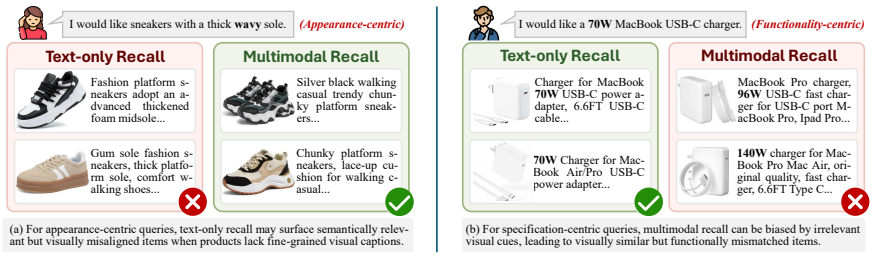


Figure 1: **The effectiveness of visual information is not universal.** For example: (a) for appearance-centric queries, visuals could capture details that are missing from text; (b) for functionality-centric queries, irrelevant visual cues may introduce noisy features and lead to suboptimal results.

and visual information [22], they typically treat modality integration as static: textual and visual signals are fused through fixed architectures and globally learned embeddings [39, 53, 68], or through predefined retrieval and matching pipelines [5, 64]. Such designs implicitly assume that the reliance on different modalities remains stable across recommendation scenarios, whereas, as the examples above illustrate, it can vary substantially across queries [74]. Consequently, fixed modality-integration strategies may yield suboptimal performance in diverse recommendation contexts. Therefore, the key challenge in multimodal recommendation is not only to fuse textual and visual information, but also to dynamically determine the appropriate reliance on each modality for a given query.

To address this challenge, we propose **AdaM-Rec**, an LLM-based framework for **Ada**ptive **M**odality routing in multimodal **Rec**ommendation. Instead of assuming that visual signals are always beneficial, AdaM-Rec adaptively calibrates the reliance on textual and multimodal evidence for user-specific queries. Its core insight is to estimate modality reliability using proxy recall tasks built from the user's historical interactions. Specifically, since the real target of an actual query is unknown, AdaM-Rec treats the user's positively interacted items as verifiable proxy targets that capture recurring user preferences. It then generates pseudo-queries that match the granularity of the actual query while pointing to these known proxy targets. By comparing the effectiveness of text-based and multimodal recall on proxy targets, AdaM-Rec iteratively estimates which modality better captures the user's interests and the actual query intent, thereby optimizing the routing strategy in an agentic manner.

This routing mechanism is built on structured representations of items, users, and their collaborative relations. Specifically, AdaM-Rec structures multimodal item information and user interaction histories into natural-language item profiles and user preferences, and models collaborative relations via similarities among user preferences. Such representations enable AdaM-Rec to agentically optimize the modality routing strategy through the native reasoning capabilities of its LLM policy. It then performs routed recall with the optimized strategy, enriches the results with collaborative items, and ranks the final candidates by their relevance to both the query and user preferences. Collectively, AdaM-Rec achieves better personalized recommendations through user-aware and query-specific routing of modality reliance. Our contributions are summarized as follows:

- We propose AdaM-Rec, to the best of our knowledge, the first framework for adaptive modality routing in multimodal recommendation. By formulating candidate recall as an agentic self-optimization process, it dynamically calibrates the reliance on textual and multimodal evidence for user-specific queries, leading to improved downstream recommendation performance.
- We introduce a novel pseudo-query mechanism for adaptive modality routing. For an actual query, AdaM-Rec generates granularity-matched pseudo-queries over user historical interactions, constructing proxy recall tasks with verifiable historical targets. This allows it to agentically determine the optimal reliance on different modalities before executing the actual query.
- Extensive experiments demonstrate that AdaM-Rec achieves strong performance compared to state-of-the-art baselines, illustrating the effectiveness and future promise of dynamic routing of modality reliance in multimodal recommender systems.

2 Related Works

2.1 LLM-based and Agentic Recommendation

In recent years, LLMs have garnered increasing attention in recommender systems due to their strong contextual modeling and reasoning capabilities [17, 72, 77]. Existing studies have integrated LLMs

into classic recommendation pipelines [33, 73], either as backbones for sequential recommendation via next-ID prediction [26, 27, 34, 75], or as auxiliary modules [16, 32] to improve collaborative filtering through representation alignment [12, 29] and knowledge/semantic enhancement [38, 45]. However, recent studies [11, 50, 71] show that item/semantic IDs and collaborative filtering representations in traditional recommender pipelines are not naturally aligned with the language space of LLMs. As a result, recommender systems may not fully exploit LLMs' language priors for complex reasoning [10, 44, 62]. This has motivated LLM-based agents [51, 55, 56, 60, 69] that formulate recommendation as an adaptive natural-language decision-making process [3, 6, 18, 20, 21, 31], enabling fine-grained preference modeling through multi-step reasoning and collaboration [36, 43, 58, 63, 66]. Inspired by the capacity of agents for context-aware decision-making and iterative optimization, we cast multimodal integration as an agentic modality-routing problem, enabling dynamic calibration of recall strategies across modalities for each user-specific query.

2.2 Multimodal Recommendation

To capture richer item features and user preferences, prior work has incorporated multimodal signals, such as images and text, into recommendation models via contrastive learning [68], multimodal tokenization [67], or adaptive feature fusion [13]. These methods enhance user and item representations beyond purely textual or ID-based signals, demonstrating the value of multimodal information for downstream recommendation tasks [22, 64]. More recently, MLLMs have shown promise in recommendation [5, 14, 47], owing to their enhanced visual understanding and inherited reasoning abilities from LLMs. It makes them a natural fit for multimodal item profiling and user modeling [42, 46, 48]. For example, MLLM-MSR [64] explores a multimodal sequential recommendation framework that leverages MLLMs to integrate product's visual and textual information and model user preferences, whereas MLLMRec [5] introduces MLLMs into collaborative modeling to improve multimodal recommendation performance. However, as noted earlier, existing approaches still lack explicit mechanisms to adaptively calibrate the use of multimodal information across different recommendation requests. In this paper, we propose an agentic framework with an adaptive modality routing mechanism that enables the recommender to dynamically regulate the importance of different modalities across recommendation contexts.

3 AdaM-Rec

AdaM-Rec is a multimodal recommendation framework equipped with an *adaptive modality routing mechanism*. Given a user-specific query, instead of assuming that visual information is always beneficial for downstream recommendation, AdaM-Rec dynamically determines the relative reliance on text-based and multimodal recall tools. As illustrated in Figure 2, the framework consists of three modules: (1) **Multimodal Item Profiling and User Modeling** (Offline), which constructs structured representations for multimodal item profiles, user preferences, and collaborative knowledge; (2) **Adaptive Modality Routing**, which agentically optimizes the allocation of the recall budget between textual and multimodal branches based on the user's query, historical interactions, and collaborative knowledge from users with similar preferences; and (3) **Modality-routed Recommendation**, which recalls candidates under the optimized modality routing schedule, augments them with collaborative candidates, and scores them according to user-specific query and preference relevance to rank the final recommendation list. Collectively, these modules enable an agentic multimodal recommendation framework that adaptively calibrates the reliance on text-based and multimodal recall according to user-specific query, historical interactions, and collaborative knowledge, thereby improving the quality of downstream personalized recommendation.

3.1 Multimodal Item Profiling and User Modeling

This module aims to construct structured profiles of multimodal item attributes, infers user preferences from historical interactions, and organizes collaborative knowledge from users with similar preference, thereby providing a foundation for downstream modality routing and personalized recommendation.

Multimodal Profiling for Items. For each item $i = (v_i, l_i)$ in raw item corpus $\mathcal{I}$, AdaM-Rec calls an MLLM ϕ_{M} with a profiling prompt $\mathcal{P}_{\text{profile}}$ to extract a structured textual item profile $s_i = \phi_{\text{M}}(i, \mathcal{P}_{\text{profile}})$, which summarizes the multimodal attributes from its textual descriptions l_i and main image v_i. It further computes a text embedding $e_i^{\text{txt}} = E_{\text{txt}}(l_i)$ and a multimodal embedding

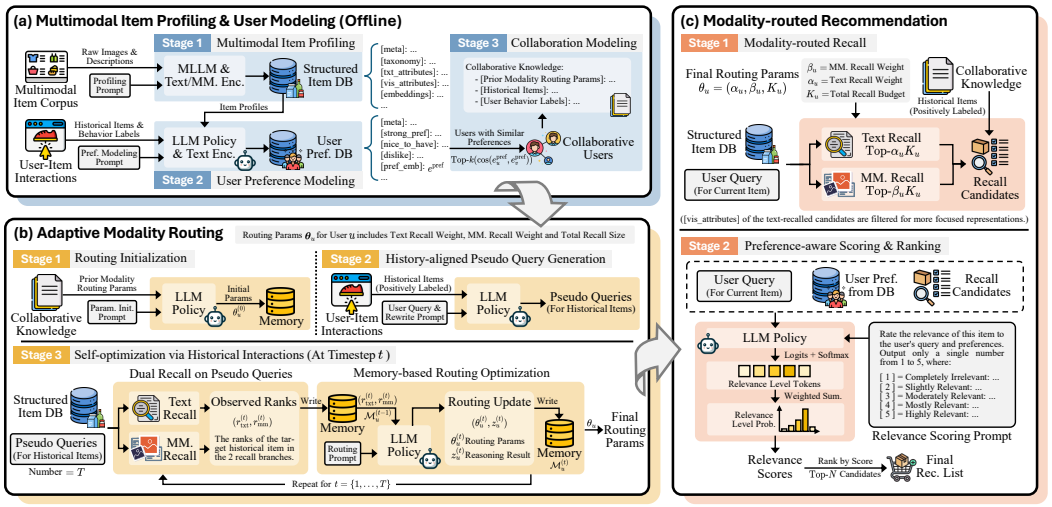


Figure 2: **Overview of AdaM-Rec.** The framework consists of three components: (a) multimodal item profiling & user modeling module, which converts raw images, textual descriptions, and user interaction histories into structured item/preference profiles and collaborative knowledge; (b) adaptive modality routing module, which operates in an agentic manner. It initializes routing from collaborative knowledge and iteratively refines the allocation of text/multimodal recall budget via history-aligned pseudo-queries and memory-based self-optimization; (c) modality-routed recommendation module, which performs text/multimodal recall under the optimized routing strategy, augments candidate items with collaborative knowledge, and ranks them by their relevance to the user's query and preferences to generate the final recommendation list.

$e_i^{\text{mm}} = E_{\text{mm}}(v_i, l_i)$ for each item via text and multimodal encoders, E_{txt} and E_{mm}, respectively. These representations form the multimodal item database:

$$\mathcal{D}^{\text{item}} = \{(s_i, e_i^{\text{txt}}, e_i^{\text{mm}})\}_{i \in \mathcal{I}}. \tag{1}$$

Modeling User Preferences from Historical Interactions. Given a user's historical interaction sequence $u = \{(i_u^t, b_u^t)\}_{t=1}^{T_u} \in \mathcal{U}$ with a length of T_u, where $b_u^t \in \{\texttt{pos}, \texttt{neg}\}$ denotes the behavior label at step t, each interacted item i_u^t is mapped to its corresponding item profile s_u^t in $\mathcal{D}^{\text{item}}$ and construct the user historical behavior profile $h_u = \{(s_u^t, b_u^t, t)\}_{t=1}^{T_u}$. AdaM-Rec then leverages its LLM policy ϕ_{L} to infer a structured textual preference profile $p_u = \phi_{\text{L}}(h_u, \mathcal{P}_{\text{pref}})$ with a preference modeling prompt $\mathcal{P}_{\text{pref}}$. The resulting profile summarizes recurring preference patterns across multimodal attributes such as style, color, material, silhouette, function, brand, compatibility and other item-level signals. Specifically, we categorize user preferences into strong positive requirements (`strong_pref`), weaker preference cues (`nice_to_have`), explicit negative constraints (`dislike`). After inferring p_u from h_u, AdaM-Rec writes them into the global user preference profile database:

$$\mathcal{D}^{\text{pref}} = \{h_u, p_u, e_u^{\text{pref}}\}_{u \in \mathcal{U}}, \tag{2}$$

where $e_u^{\text{pref}} = E_{\text{txt}}(p_u)$ is the textual embedding of the user's preference profile, thereby incorporating the current user into a retrievable collaborative space for other users.

Preference-based Collaboration Modeling. To model collaborative relations among users, AdaM-Rec first computes the cosine similarity between the current user's preference embedding e_u^{pref} and those of others in $\mathcal{D}^{\text{pref}}$. It then retrieves the top-K_c most similar users as the collaborative user set $\mathcal{N}_u$. For each collaborative user $v \in \mathcal{N}_u$, the collaborative knowledge consists of (θ_v, s_v, b_v). Specifically, θ_v denotes the previously optimized modality routing parameters of user v (Sec. 3.2), if available. Meanwhile, s_v and b_v denote the historically interacted items of user v and the corresponding behavior labels, respectively. The collaborative knowledge for the current user can be represented as:

$$\mathcal{C}_u = \{(\theta_v, s_v, b_v)\}_{v \in \mathcal{N}_u}. \tag{3}$$

3.2 Adaptive Modality Routing

This module optimizes the modality-routing strategy for user-specific recommendation requests by considering the current actual query, historical interactions, and collaborative knowledge. It

introduces a novel routing mechanism that agentically constructs proxy recall tasks through pseudo-queries aligned with the user's historically interacted targets and the granularity of the actual query. By evaluating text-based and multimodal recall branches on these proxy tasks, it adaptively allocates modality weights and recall budget at inference time for the actual query.

Initialization from Collaborative Priors. Given the collaborative knowledge $\mathcal{C}_u$ of the current user, the LLM policy ϕ_{L} leverages the parameter-initialization prompt $\mathcal{P}_{\mathrm{pi}}$ to analyze common priors in the modality-routing parameters of collaborative users, thereby estimating the initial routing state $\theta_u^{(0)}$ for the user:

$$\theta_u^{(0)} = \left(\alpha_u^{(0)}, \beta_u^{(0)}, K_u^{(0)}\right) = \phi_{\mathrm{L}}(\mathcal{C}_u, \mathcal{P}_{\mathrm{pi}}), \tag{4}$$

where $\alpha_u^{(0)}$ and $\beta_u^{(0)}$ denote the weights for text-based and multimodal recall branches at timestep $t = 0$, respectively, while $K_u^{(0)}$ denotes the initial recall budget. The weights satisfy $\alpha_u^{(0)} + \beta_u^{(0)} = 1$. Subsequently, this initial state is stored in memory $\mathcal{M}_u^{(0)} = \{\theta_u^{(0)}\}$ to facilitate subsequent iterative optimization, where K_u is constrained to be no greater than a threshold constant τ.

Self-optimization via Historical Interactions. To make modality routing personalized rather than purely heuristic, AdaM-Rec optimizes routing parameters based on current user's own historical interacted items. The key motivation is that, for a user's actual query, the target item is unknown, making it difficult to directly determine the effectiveness of text-based and multimodal recall for the query. In contrast, related and positively interacted historical items are observed choices from the same user and thus provide user-specific, verifiable proxy targets that reflect the user's recurring preference patterns, facilitating the optimization of routing parameters.

Specifically, given the current actual query q, the LLM policy ϕ_{L} uses a rewriting prompt $\mathcal{P}_{\mathrm{rw}}$ to reformulate it into a set of history-aligned pseudo-queries $\widetilde{\mathcal{Q}}_u = \phi_{\mathrm{L}}(q, h_u, \mathcal{P}_{\mathrm{rw}}) = \{\tilde{q}_1, \tilde{q}_2, \ldots, \tilde{q}_{T_f}\}$, where each pseudo-query $\tilde{q}_t$ is expressed at a granularity similar to q but targets one known and related historical item from the user's interaction history. During this process, ϕ_{L} also filters out historical items that are irrelevant with respect to q or negatively interacted, resulting in a filtered interaction sequence of length T_f. In this way, AdaM-Rec constructs proxy retrieval tasks that preserve the search form of the current recommendation request while replacing unknown target with verifiable user-preferred items. Subsequently, it iteratively optimizes the recall strategy based on these pseudo-queries and history items. At timestep $t > 0$, it invokes the text-based and multimodal recall on $\tilde{q}_t$, and obtains the rank positions of the target item i_u^t in the corresponding retrieval results, denoted by $r_{\text{txt}}^{(t)}$ and $r_{\text{mm}}^{(t)}$, respectively. These relative ranks provide evidence about which modality better matches the user's preference structure under analogous search conditions. Based on these observations, the previous status stored in the memory $\mathcal{M}_u^{(t-1)}$, and the modality-routing prompt $\mathcal{P}_{\mathrm{mr}}$ that guides the reasoning process, the LLM policy ϕ_{L} infers a refined parameter state for the next iteration:

$$\left(\theta_u^{(t)}, z_u^{(t)}\right) = \phi_{\mathrm{L}}\left(r_{\text{txt}}^{(t)}, r_{\text{mm}}^{(t)}, \mathcal{M}_u^{(t-1)}, \mathcal{P}_{\mathrm{mr}}\right), \quad T_f \geq t > 0, \tag{5}$$

where $z_u^{(t)}$ denotes the policy's reasoning summary for the update. To preserve the optimization trajectory, the result of each iteration is written into the user-specific memory:

$$\mathcal{M}_u^{(t)} = \mathcal{M}_u^{(t-1)} \cup \left\{\left(r_{\text{txt}}^{(t)},\, r_{\text{mm}}^{(t)},\, \theta_u^{(t)},\, z_u^{(t)}\right)\right\}. \tag{6}$$

After processing all pseudo-queries, the final modality routing parameters for the current user and query are taken as $\theta_u = \theta_u^{(T)}$, which are then used to control downstream recall for the real query. Moreover, the routing parameter θ_u is written back into the current user's preference profile $\mathcal{D}^{\mathrm{pref}}[u] \leftarrow \{\theta_u\}$, enabling its reuse as collaborative knowledge for other users.

3.3 Modality-Routed Recommendation

This module performs downstream retrieval using the final modality routing parameters optimized in the previous stage. It allocates recall budget across text-based and multimodal branches, augments the recalled candidates with collaborative evidence from preference-similar users, and applies preference-aware scoring to rank a recommendation list aligned with both query intent and the user's preferences.

Modality-Routed Recall. Given the query q and the item database $\mathcal{D}^{\text{item}} = \{(s_i, e_i^{\text{txt}}, e_i^{\text{mm}})\}_{i \in \mathcal{I}}$, AdaM-Rec invokes downstream text-based and multimodal recall with the optimized final routing parameters $\theta_u = (\alpha_u, \beta_u, K_u)$ to construct an initial recall set $\mathcal{R}_{u,q}^{\text{init}}$, by taking the Top-$\alpha_u K_u$ items whose e_i^{txt} are most similar to query's embedding $E_{\text{txt}}(q)$ in textual space, and the Top-$\beta_u K_u$ items

whose e_i^{mm} are most similar to query's embedding $E_{\text{mm}}(q)$ in multimodal space. For candidates recalled by text-based retrieval, visual attributes in their profiles are selectively filtered to produce cleaner and more focused candidate representations.

Collaborative Recall Augmentation. To complement the recall results, we further incorporate positive historical items from preference-similar users as collaborative candidates:

$$\mathcal{R}_u^{\text{co}} = \left\{ s_v^t \;\middle|\; v \in \mathcal{N}_u,\ 1 \le t \le T_v,\ b_v^t = \texttt{pos} \right\}, \qquad \mathcal{R}_u^{\text{aug}} = \mathcal{R}_u^{\text{init}} \cup \mathcal{R}_u^{\text{co}}. \tag{7}$$

This augmentation complements user-specific modality routing by injecting collaborative evidence from preference-similar users, thereby improving recall coverage and increasing the likelihood of retrieving items that align with the current user's latent preferences.

Preference-aware Scoring & Ranking. Given the augmented candidate set, AdaM-Rec estimates a preference-aware relevance score for each candidate. Conditioned on the query q, the user preference profile p_u, the candidate profile s_i, and the scoring prompt $\mathcal{P}_{\text{sc}}$, the LLM policy ϕ_{L} outputs a softmax-normalized probability distribution over discrete relevance-level tokens $\ell \in \{1, 2, 3, 4, 5\}$:

$$P_{\text{R}}(\ell \mid q, p_u, s_i, \mathcal{P}_{\text{sc}}) = \frac{\exp(\phi_{\text{L}}(\ell \mid q, p_u, s_i, \mathcal{P}_{\text{sc}}))}{\sum_{k=1}^{5} \exp(\phi_{\text{L}}(k \mid q, p_u, s_i, \mathcal{P}_{\text{sc}}))}, \tag{8}$$

where $\phi_{\text{L}}(\cdot)$ denotes the logits produced by the LLM policy, and $\mathcal{P}_{\text{sc}}$ guides the policy to assess whether each candidate satisfies the query intent q while aligning with the user preferences in p_u. The final candidate score is computed as the expected relevance level:

$$Score_i = \sum_{\ell=1}^{5} \ell \cdot P_{\text{R}}(\ell \mid q, p_u, s_i, \mathcal{P}_{\text{sc}}). \tag{9}$$

This formulation assigns each candidate a continuous score based on query intent, user preferences, and item profiles. Candidates with higher scores are deemed more relevant and better aligned with user's queries and personalized preferences, with the top-N forming the final recommendation list.

4 Experiments

4.1 Dataset and Implementation

Datasets. To ensure fair comparisons with prior work, we strictly follow the experimental settings of the corresponding baselines in each recommendation task. For query-based recommendation, we use the user-query datasets constructed by TAIRA [65] on three Amazon Review subsets [8, 28]: Amazon Clothing & Shoes, Amazon Beauty, and Amazon Music. These datasets contain 94,328, 91,263, and 302,803 items, respectively, with 361, 374, and 393 carefully curated user-query pairs. For the extended preference-based recommendation, we adopt the evaluation protocol of MLLM-MSR [64] on Amazon Video Games and Amazon Baby Products. After standard 5-core preprocessing [64], the two datasets include 12,559 and 19,945 users, and 18,017 and 24,847 items, respectively.

Implementation. AdaM-Rec uses Qwen3-8B [59] as the default LLM policy. It employs Qwen3-Emb.-0.6B [70] as the text encoder E_{txt}, Qwen3-VL-Emb.-2B [24] as the multimodal encoder E_{mm}, and Qwen3-VL-8B-Instruct [1] as the MLLM ϕ_{M} for multimodal profiling. For a fair comparison, we upgrade the backbones of earlier baselines that adopt weaker (M)LLMs or encoders to the same setting as ours. All open-source models are deployed on A100 80GB GPUs. Most hyperparameters are automatically optimized by the policy, except for the collaborative users number $K_c = 5$ and maximum recall budget threshold $\tau = 1000$, following the analysis in Sec.4.4. Unless otherwise stated, we report the median over five trials, perform ablations on Amazon Beauty, and randomly sample 20% of users as the test set for preference-based recommendation. Prompts involved in Sec.3 and profile examples are provided in Appendices A and B, respectively.

4.2 Comparison Against Other Methods

Query-based Recommendation. As shown in Table 1, we evaluate AdaM-Rec on query-based personalized recommendation task, where the system recalls and ranks items from the full corpus based on the user's query and preferences inferred from historical interactions. Following TAIRA [65], we assess full-corpus groundtruth item recommendation using Hit Rate (HR) and Normalized Discounted Cumulative Gain (NDCG). We compare against the same categories of baselines as TAIRA, including classical retrieval/reranking models [4, 23, 40], general agentic frameworks [35, 61] instantiated under

Table 1: **Comparison with query-based recommendation baselines.** "@K" denotes the HR and NDCG at top-K. Bold and underlined values indicate the best and second-best results, respectively.

Method	Amazon Beauty				Amazon Clothing				Amazon Music			
	HR ↑		NDCG ↑		HR ↑		NDCG ↑		HR ↑		NDCG ↑	
	@20	@40	@20	@40	@20	@40	@20	@40	@20	@40	@20	@40
BM25 [40]	0.0348	0.0508	0.0166	0.0199	0.0194	0.0277	0.0080	0.0096	0.0382	0.0483	0.0183	0.0203
BGE-M3 [23]	0.0374	0.0561	0.0173	0.0211	0.0249	0.0388	0.0093	0.0122	0.0458	0.0611	0.0239	0.0269
BGE-Reranker [4]	0.0401	0.0615	0.0184	0.0227	0.0277	0.0443	0.0137	0.0171	0.0483	0.0687	0.0204	0.0245
ReAct [61]	0.0909	0.1096	0.0369	0.0408	0.0499	0.0720	0.0278	0.0322	0.0789	0.0967	0.0411	0.0448
Reflexion [35]	0.0963	0.1230	0.0491	0.0547	0.0582	0.0776	0.0308	0.0349	0.0840	0.0992	0.0494	0.0525
InteRecAgent [15]	0.0989	0.1257	0.0454	0.0508	0.0360	0.0693	0.0141	0.0207	0.0814	0.0916	0.0632	0.0651
MACRS [6]	0.1016	0.1310	0.0473	0.0533	0.0443	0.0554	0.0175	0.0197	0.0585	0.0687	0.0221	0.0243
MACRec [49]	0.1471	0.1765	0.0803	0.0865	0.0554	0.0665	0.0283	0.0305	0.0763	0.0891	0.0569	0.0595
MACF [57]	0.1551	0.1872	0.0725	0.0791	0.0886	0.1191	0.0510	0.0574	0.1221	0.1374	0.0679	0.0710
TAIRA [65]	0.1497	0.1925	0.0798	0.0886	0.0914	0.1274	0.0457	0.0532	0.1349	0.1450	0.0939	0.0960
AdaM-Rec	**0.1711**	**0.2193**	**0.0979**	**0.1077**	**0.1108**	**0.1468**	**0.0574**	**0.0648**	**0.1501**	**0.1629**	**0.1063**	**0.1089**

Table 2: **Comparison with preference-based recommendation baselines.** "@K" denotes HR and NDCG at top-K. Bold and underlined values indicate best and second-best results.

Method	Amazon Video Games				Amazon Baby Products			
	HR ↑		NDCG ↑		HR ↑		NDCG ↑	
	@20	@40	@20	@40	@20	@40	@20	@40
GRU4Rec [9]	0.0587	0.0865	0.0326	0.0383	0.0391	0.0704	0.0218	0.0282
SASRec [19]	0.0655	0.0976	0.0343	0.0410	0.0424	0.0797	0.0236	0.0312
MGAT [39]	0.0611	0.0921	0.0330	0.0394	0.0416	0.0762	0.0228	0.0299
MMGCN [53]	0.0647	0.1012	0.0347	0.0422	0.0429	0.0822	0.0241	0.0322
MMSR [13]	0.0694	0.1067	0.0359	0.0436	0.0547	0.0905	0.0256	0.0329
Trans2D [37]	0.0536	0.0798	0.0283	0.0337	0.0346	0.0569	0.0178	0.0224
MLLM-MSR [64]	0.0706	0.1099	0.0367	0.0478	0.0448	0.0787	0.0208	0.0277
AB-Rec [54]	0.0734	0.1179	0.0296	0.0387	0.0652	0.1073	0.0251	0.0337
LLaRa [25]	0.0786	0.1083	0.0329	0.0396	0.0531	0.0895	0.0201	0.0276
LLMRec [52]	0.0849	0.1187	0.0401	0.0473	0.0622	0.0950	0.0243	0.0315
TallRec [2]	0.0758	0.1052	0.0334	0.0394	0.0479	0.0862	0.0191	0.0270
SPRec [7]	0.0853	0.1159	0.0390	0.0452	0.0454	0.0822	0.0204	0.0279
RecZero [20]	0.0873	0.1143	0.0362	0.0421	0.0509	0.0920	0.0196	0.0286
MLLMRec-R1 [47]	0.0940	0.1226	**0.0434**	0.0481	0.0679	0.1083	0.0261	0.0344
AdaM-Rec	**0.0967**	**0.1267**	0.0422	**0.0484**	**0.0702**	**0.1138**	**0.0264**	**0.0351**

the TAIRA setting, and state-of-the-art (SOTA) interactive agentic recommender [6, 15, 49, 57, 65] compatible with this task. Results are taken from open-source codebases or reproduced using official settings. AdaM-Rec consistently outperforms prior baselines on query-based recommendation tasks, achieving average gains of 14.1% in HR and 15.9% in NDCG, demonstrating its effectiveness.

Preference-based Recommendation. As shown in Table 2, we further evaluate AdaM-Rec on the query-free preference-based sequential recommendation task, which predicts the next interacted item based solely on user preferences inferred from historical interactions. To adapt AdaM-Rec to this setting, we replace the query with the user's current preference for modality routing and downstream retrieval. For each historical item involved in modality routing, its corresponding preference is constructed using only preceding visible interactions, while strictly preserving the format and granularity of the target preference. Following the more challenging MLLM-MSR protocol, we report HR and NDCG by ranking one ground truth item against 1,000 randomly sampled items for each test instance. We compare with baselines aligned with MLLM-MSR [64], including classical and multimodal sequential recommenders [9, 13, 19, 37, 39, 53] and (M)LLM-based methods [2, 7, 20, 25, 47, 52, 54, 64]. Results are obtained from open-source codebases. Although this task is not the primary query-based recommendation scenario targeted by AdaM-Rec, it still achieves competitive performance against SOTA baselines, further demonstrating its generalizability.

4.3 Ablation Studies

LLM Policy. The first group in Table 3 shows that stronger policy improve modality routing and preference judgment, leading to better recall and ranking performance, though with diminishing gains. Therefore, we adopt Qwen3-8B as the default policy for its cost-performance balance.

Adaptive Routing Mechanism. The second group in Table 3 examines the routing-related components. "w/o Adaptive Modality Routing" replaces adaptive routing with the optimal static parameters

Table 3: **Ablation studies on agent policy and core components.** The metric "Recall" denotes the HR of candidates recalled with routed parameters. "@K" denotes the HR and NDCG scores computed over top-K ranked items. Bold highlights the default config and the strongest policy variant.

Method	Recall ↑	HR ↑		NDCG ↑	
		@20	@40	@20	@40
Full (Qwen3-8B policy)	**0.4278**	**0.1711**	**0.2193**	**0.0979**	**0.1077**
LLM Policy					
Qwen3-32B	**0.4358**	**0.1765**	**0.2246**	**0.1017**	**0.1116**
Adaptive Routing Mechanism					
w/o Adaptive Modality Routing	0.3743	0.1524	0.1898	0.0787	0.0866
w/o Collaborative Priors Initialization	0.4171	0.1684	0.2139	0.0969	0.1065
w/o Pseudo-query & Self-optimization	0.3850	0.1578	0.1979	0.0904	0.0988
w/o Memory	0.3930	0.1631	0.2032	0.0938	0.1022
Candidate Recall					
w/o Collaborative Recall Augmentation	0.4091	0.1658	0.2086	0.0964	0.1054
w/o Text-based Recall	0.3155	0.1390	0.1631	0.0726	0.0774
w/o Multimodal Recall	0.3422	0.1471	0.1765	0.0768	0.0829
Preference-aware Ranking					
w/o Attribute Filtering	0.4278	0.1551	0.2059	0.0743	0.0848
w/o Preference-aware Scoring	0.4278	0.1337	0.1684	0.0576	0.0647

Table 4: **Ablations of embedding backbones within different recall branches.** "Recall" denotes HR of candidates recalled with routed parameters. "@K" denotes the HR and NDCG at top-K.

Text Encoder	Multimodal Encoder	Recall ↑	HR ↑		NDCG ↑	
			@20	@40	@20	@40
SigLIP2-L/16	SigLIP2-L/16	0.3289	0.1417	0.1711	0.0745	0.0809
	Qwen3-VL-Emb.-2B	0.3904	0.1631	0.2032	0.0865	0.0952
	Qwen3-VL-Emb.-8B	0.4091	0.1684	0.2112	0.0879	0.0972
Qwen3-Emb.-0.6B	SigLIP2-L/16	0.3449	0.1471	0.1791	0.0815	0.0884
	Qwen3-VL-Emb.-2B	0.4278	0.1711	0.2193	0.0979	0.1077
	Qwen3-VL-Emb.-8B	0.4412	0.1738	0.2246	0.0972	0.1083
Qwen3-Emb.-4B	SigLIP2-L/16	0.3583	0.1497	0.1845	0.0828	0.0904
	Qwen3-VL-Emb.-2B	0.4331	0.1711	0.2219	0.0971	0.1082
	Qwen3-VL-Emb.-8B	0.4492	0.1764	0.2273	0.0981	0.1091

in Figure 3, where the text-based/multimodal recall weights are both 0.5, and the total budget is 800, causing a clear performance drop. This highlights the importance of dynamically routing modality reliance across user-specific queries. "w/o Collaborative Priors Initialization" removes collaborative-prior-based routing initialization and uses the optimal parameters of static strategy as initial state instead, leading to slight degradation, which verifies its effectiveness as warm start. "w/o Pseudo-query & Self-optimization" keeps only collaborative initialization for each user and shows a larger decline, demonstrating the importance of pseudo-query-based self-optimization for adapting routing parameters to actual query intent. "w/o Memory" further degrades performance, confirming that memory helps stabilize and improve self-optimization by reusing useful past routing experiences.

Candidate Recall. The third group in Table 3 evaluates the recall-stage design. "w/o Collaborative Recall Augmentation" removes historical items from collaborative users, reducing both candidate recall and downstream ranking performance, showing that collaborative knowledge could improve candidate coverage. "w/o Text-based/Multimodal Recall" uses only one recall branch with the same total budget as static setting above, which also makes adaptive modality routing invalid. Their substantial degradation indicates the complementarity and necessity of text-based/multimodal recall. The larger drop from removing text-based recall suggests that it may act as the primary semantic anchor, while multimodal recall provides additional cues that cannot be fully captured by text alone.

Preference-aware Ranking. The last group in Table 3 analyzes the ranking-stage components. "w/o Attribute Filtering" removes visual-attribute filtering from text-recalled candidate profiles, leading to lower downstream HR and NDCG and validating its role in maintaining cleaner, task-relevant representations for improving preference-aware ranking. "w/o Preference-aware Scoring" adopts recall-stage embedding similarity for downstream ranking and causes a drop on ranking performance. This confirms that effective recommendation depends not only on retrieving relevant candidates, but also on accurately ranking them according to the current query intent and user preferences.

Embedding backbones. As shown in Table 4, we conduct ablations on the textual and multimodal embedding backbones used for recall. The results show that stronger encoders generally improve

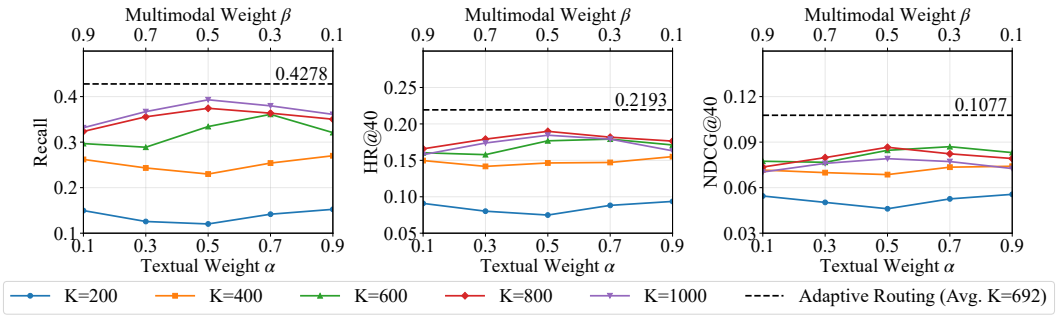


Figure 3: **Comparison of static/adaptive modality routing.** K denotes total recall budget, while α/β denote text-based/multimodal recall weights, respectively. "@40" denotes HR/NDCG at top-40.

candidate recall, but the gains do not scale proportionally. Scaling up the multimodal encoder brings slightly larger improvements than enlarging the text encoder, suggesting that a compact text encoder is sufficient for text-based recall in our setting, while multimodal recall is more sensitive to encoder capacity. Moreover, using SigLIP2 [41] as a vision-only alternative to the multimodal encoder leads to a clear performance drop compared with vision-language encoders such as Qwen3-VL-Embedding. This suggests that the recall benefits from combining visual cues with language-based semantics, such as function, brand, and other fine-grained purchase constraints [76], rather than relying on visual features alone. Overall, we adopt Qwen3-Emb.-0.6B and Qwen3-VL-Emb.-2B as the default configuration for a better efficiency-performance trade-off.

4.4 Discussion

Static and Adaptive Modality Routing. We analyze different recall budgets by varying the text-based recall weight and assigning the remaining weight to the multimodal branch. As shown in Figure 3, adaptive routing consistently outperforms static strategies, confirming the benefit of dynamically calibrating modality reliance and recall allocation for a specific query. Under fixed routing, when the recall budget is small, concentrating it on the more suitable modality can be more effective than evenly splitting it across branches, since each branch may otherwise receive too few candidates for reliable retrieval. As the budget increases, however, retrieving more candidates also increases the burden on downstream ranking, which may hinder final recommendation quality. These results suggest that fixed routing is difficult to generalize across recommendation requests, highlighting the need for adaptive per-query modality allocation.

Limitations. Despite its superior performance, AdaM-Rec introduces additional computational overhead due to its agentic workflow. This reflects a common trade-off in agentic recommender systems, where more adaptive decision-making can improve recommendation quality at the cost of increased inference complexity. Therefore, making the inference process more efficient without compromising recommendation performance remains an important direction for future work.

5 Conclusion

We propose AdaM-Rec, to the best of our knowledge, one of the first systematic efforts to conceptually integrate adaptive modality routing into multimodal recommendation under an agentic paradigm. Since different query intents exhibit varying degrees of reliance on multimodal features and textual semantics, indiscriminate modal fusion may introduce noisy cues and impair recommendation quality. To address this challenge, AdaM-Rec introduces an innovative adaptive modality routing mechanism. Specifically, the agent generates pseudo-queries that align with the information granularity of the current query but target known items from the user's historical interactions. By simulating verifiable proxy recall tasks with pseudo-queries to perform self-optimization, the framework agentically allocates the weights and budgets between the text-based and multimodal recall branches for the actual query at inference time. Comprehensive evaluations show that AdaM-Rec achieves highly competitive results against SOTA methods, providing evidence that adaptively routing modality reliance is a worthwhile exploration for improving multimodal recommender systems.

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A Prompts

Multimodal Item Profiling Prompt

You are an expert e-commerce item profiler. Given product text and its main image, extract a fine-grained feature profile in STRICT JSON.

Item text fields: title: {item.title}, detail_text: {item.detail_description}, category_hint: {item.category_hint or 'unknown'}.

Item profiling requirements:

1) Type-first taxonomy:

- Always output ‘item_type‘ (required).
- If hierarchical category is uncertain, keep only ‘item_type‘ and leave ‘category_path‘ empty.
- If known, output ‘category_path‘ as a list (e.g., ["Electronics", "Gaming", "Headset"]).
- Also infer use_case, target_people, seasonality.

2) Textual attribute tags (fine-grained):

- title keyword summary (must leverage title)
- material/fabric composition
- core features & specs (size, capacity, weight, dimensions, compatibility, power, ingredients, etc.)
- package/bundle information
- quality & durability claims
- comfort/usability claims
- price, price_band inference (budget/mid/premium) and value_for_money signal

3) Visual attribute tags (from image):

- dominant colors (+ optional hex-like names)
- silhouette/shape/form factor
- style keywords (minimalist, sporty, retro, luxury, kawaii, etc.)
- texture/finish (matte/glossy/metallic/knit/grainy)
- pattern/print/logo density
- scene mood (formal, youthful, homey, professional, outdoor, gaming, etc.)
- perceived quality level (low/medium/high with confidence)

4) Output ONLY one JSON object. No markdown.

JSON schema:

{ "item_id": "{item.item_id}", "title": "{item.title}", "taxonomy": { "item_type": "", "category_path": [], "use_case": [], "target_people": [], "seasonality": "" }, "text_attributes": {...}, "visual_attributes": {...} }

Preference Modeling Prompt

You are a user preference modeling expert for an e-commerce recommendation system.

Task: Based on the user's relevant historical positive and negative behaviors, reason through the user's current preferences and shopping intent in chronological order.

Requirements:

1) Do not assume or reference query information that does not exist.

2) Clearly distinguish Strong_Preference / Nice_to_Have / Dislike.

3) Preferences inferred from the positive sequence in the user history, especially positive behaviors, should be prioritized into Strong_Preference or Nice_to_Have.

4) If the history contains analyzable visual information, such as visual_tags or image-derived descriptions, Nice_to_Have must include conclusions about visual preferences. If there is no analyzable visual information, do not reference or fabricate non-existent visual information.

5) The history is already provided in ascending timestamp order: positive behaviors should be analyzed chronologically to infer evolving preferences and purchase trajectory; negative samples do not need to be reasoned about in chronological order.

6) Reasoning must combine contrastive evidence from both positive and negative behaviors in the history.

7) Strong_Preference and Nice_to_Have do not have a fixed number of items; decide the number of items based on the strength of the evidence.

8) Every element in Strong_Preference, Nice_to_Have, and Dislike must be a short natural-language preference phrase, such as “Nintendo Switch compatibility” or “wireless/Bluetooth connection”. Do not output JSON fragments, dictionaries, key-value pairs, or quoted structured snippets as elements.

9) Preconditions must take priority over general preferences. Do not output conclusions that conflict with the preconditions. The preconditions are as follows: {preconditions_prompt}.

10) Output a strict JSON object with the following fields: Strong_Preference, Nice_to_Have, Dislike, Reasoning_Process.

Relevant historical records (JSON): {json.dumps(history_interactions_with_item_profiles, ensure_ascii=False)}

Preconditions Prompt

Preconditions:

A) First determine the product type of the user's current intent and the key preconditions before outputting preferences.

B) Any attribute that conflicts with the preconditions must be placed in Dislike, and conflicting items must not appear in Strong_Preference or Nice_to_Have.

C) If a user query exists and the historical behaviors conflict with the current query, prioritize the preconditions from the current query; the history should only be used as supplemental evidence for style, budget, theme, or similar preferences.

D) Do not recommend cross-platform or incompatible products, such as PC games vs. PS/Xbox/Switch games, or iOS accessories vs. Android-only accessories.

E) For audience-sensitive categories, the target-audience preconditions must be checked, such as gender, age group, and sizing system.

F) For technical products, compatibility preconditions must be checked, such as system version, interface/protocol, power/voltage, and size specifications.

G) If there is insufficient information to confirm compatibility, explicitly include "avoid incompatible/platform-mismatched products" in Dislike, and explain the uncertainty in the Reasoning Process.

H) Before outputting, perform a consistency self-check: Strong_Preference and Dislike must not contradict each other, and all Strong_Preference items must satisfy the preconditions.

Param Initialization Prompt

Please identify the common pattern across the similar users' parameter results, and then generate the initialization parameters for a new user based on that common pattern. The input contains only three types of parameters:

- text_weight: text weight

- vl_weight: visual weight

- recall_size: total recall budget

Requirements:

1) Output a consensus parameter triplet: text_weight, vl_weight, recall_size.

2) text_weight + vl_weight must equal 1, with a rounding tolerance of ±0.001.

3) Both text_weight and vl_weight must be within [0.30, 0.70]. As an initialization strategy, start from a balanced and conservative setting.

4) recall_size must be within {max_recall_size} and must be an integer.

5) If these users show large disagreement, prioritize a robust middle value and do not let extreme values dominate.

6) Also include a reasoning field that briefly explains the common pattern among the similar users and why the final values were chosen.

7) Output only one JSON object, without any other explanatory text.

Input data: {collaborative_users_params}

Strict output format:

{"text_weight": "", "vl_weight": "", "recall_size": "", "reasoning": ""}

Pseudo Query Rewrite Prompt

Given an actual query and the profile of a history item, please generate a pseudo query that satisfies the following requirements:

1) The information granularity must be consistent with the actual query, neither broader nor more specific.

2) The semantic slot structure should be as consistent as possible, such as category, brand, specification, function, target audience, style, scenario, budget, and similar slots.

3) The specific content must point to the given history item and must be supported by its title, category, or attributes. Do not fabricate information.

4) Do not rewrite the history item into a full long description. Output only a search query that a user might type.

5) If the actual query contains hard constraints, such as brand, model, size, compatibility, or target audience, map them only when the history item provides corresponding evidence. If there is no evidence, keep the expression weakened at the same level of granularity and do not make things up.

6) Output only one line of pseudo query text, with no explanation.

7) If the content of the given historical item clearly mismatches the potential target of the actual query, **must** ignore the above conditions and directly output only "INVALID".

Input:

Actual query: {actual_query}, History item information (JSON): {history_item_json}

Please output: <one-line pseudo query only>

Modality Routing Prompt

You are a modality-routing controller for an adaptive multimodal recommendation system.

Context: A history-aligned pseudo-query was generated for an item the user previously liked. Both text-based recall and vision-language recall were run on this pseudo-query. Since the target item is known, text_rank and vl_rank show which modality is more reliable for the current recommendation intent.

Task: Update the routing parameters for the next iteration: (1) text_weight: reliance on text-based semantic recall. (2) vl_weight: reliance on visual or vision-language similarity recall. (3) total_recall: overall recall budget.

Goal: Optimize text_weight, vl_weight, and total_recall based on the pseudo-query, target item profile, text_rank, vl_rank, current routing state, memory, and constraints.

Routing principles: (1) Lower rank is better. (2) Strongly favor the modality that retrieves the target much better or retrieves it while the other misses. (3) Prefer differentiated weights when evidence is clear. (5) Use moderate weights such as 0.65/0.35 or 0.7/0.3 when evidence is suggestive but not decisive. (6) Use balanced weights only when evidence is genuinely mixed or weak. (7) Must prioritize consistent memory trends over a single noisy observation. (8) If current evidence conflicts with memory, update conservatively unless the new evidence is very strong.

Text-dominant signals: Increase text_weight when the evidence is mainly semantic, functional, or specification-driven, such as brand, model, compatibility, product type, function, material, size, capacity, wattage, voltage, flavor, scent, ingredient, quantity, title, or version. If text recall clearly beats VL recall, strongly favor text.

VL-dominant signals: Increase vl_weight when the evidence depends mainly on visual appearance, such as style, color, shape, silhouette, texture, pattern, decoration, aesthetic, design, look, packaging, or visual similarity. If VL recall clearly beats text recall, strongly favor VL.

Mixed guidance: Use moderate or balanced weights when textual constraints and visual preferences are both important, ranks are close, wins alternate, or modalities provide complementary evidence. Prefer mild differentiation such as 0.6/0.4 or 0.7/0.3 over exactly 0.5/0.5 when there is any clear tendency.

Recall-size guidance: (1) Increase total_recall if both modalities miss, ranks are poor, or the query is broad or ambiguous. (2) Decrease total_recall if the dominant modality retrieves targets reliably at high ranks and the query is specific. (3) Keep total_recall stable when evidence is mixed or the previous value seems adequate. (4) Do not increase recall only to compensate for a bad modality; reduce that modality's weight when confidence is high.

Constraints: (1) text_weight and vl_weight must be floats in [0.0, 1.0]. (2) text_weight + vl_weight should be 1.0. (3) total_recall must be a positive integer. (4) Obey input bounds, especially max recall size max_recall_size and min recall size min_recall_size.

Output: Return strictly one valid JSON object only, with no markdown or extra text.

JSON schema: { "text_weight": float, "vl_weight": float, "total_recall": int, "summary": string, "reasoning": string }

The reasoning field should briefly state which modality performed better, whether the query is text-driven, VL-driven, or mixed, how memory/rule_hints affected the update, and why total_recall changed or stayed stable.

Input:

{ "step": ..., "target_item_profile": {history_item_profile}, "pseudo_query": {pseudo_query}, "observation": { "text_rank": ..., "vl_rank": ... } "current_state": ..., "memory": ... "constraints": { "max_recall_size": ..., "min_recall_size": ... } }

Relevance Scoring Prompt

You are an expert in fine-ranking e-commerce recommendations. Please judge, from the user's perspective, how well the candidate product matches the user's current preferences, and give a score from 1 to 5.

Scoring rules:

[1] = Completely Irrelevant. Clearly violates Dislike, or obviously conflicts with the core intent of the user's query, and should not be recommended even if it satisfies some preferences.

[2] = Slightly Relevant. Only weakly related to the query, matching only a few superficial keywords or marginal attributes, and does not effectively satisfy the user's current core needs; or satisfies very few Strong_Preference requirements.

[3] = Moderately Relevant, satisfies part of the query's core needs, or satisfies some Strong_Preference / multiple Nice_to_Have requirements, but still has obvious gaps in product type, key attributes, or usage scenario.

[4] = Mostly Relevant, satisfies the query's core needs well, covers most Strong_Preference requirements, and gains some bonus from Nice_to_Have, with no obvious conflicts.

[5] = Highly Relevant, fully satisfies the query's core needs and Strong_Preference requirements, has no Dislike conflicts, and stands out in Nice_to_Have, Predicted_Next_Items, or the user's potential preferences.

Input information:

- User Query: {query}
- Strong_Preference: {strong_pref}
- Nice_to_Have: {nice_to_have}
- Dislike: {dislike}
- Candidate profile: {item_profile}

Output requirements: Output only one number: 1/2/3/4/5. Do not output explanations, punctuation, spaces, or any other content.

B Profile Formats

Multimodal Item Profile Format Example

{
"item_id": ...,
"taxonomy": {"item_type": ..., "category_path": [...], "use_case": [...], "target_people": [...], "seasonality": ..., ...},
"text_attributes": {"title_keyword_summary": [...],"material_fabric_composition": [...], "core_features_specs": {"size": ..., "capacity": ..., "weight": ..., "dimensions": ..., "compatibility": ..., "power": ..., "ingredients": ..., "package_bundle_information": [...], "quality_durability_claims": [...], "comfort_usability_claims": [....], "price": ..., "price_band_inference": ..., "value_for_money_signal": ..., ...}, ...},
"visual_attributes": {"dominant_colors": [...], "silhouette_shape_form_factor": ..., "style_keywords": [...], "texture_finish": ..., "pattern_print_logo_density": ..., "scene_mood": ..., "perceived_quality_level": ..., ...}
}

User Preference Profile Format Example

{
"user_id": ...,
"preferences": { "strong_pref": [...], "nice_to_have": [...], "dislike": [...], "reasoning": ... },
"history_meta": {"historical_item_profiles": {...}, "behavior_labels": {...}, "timestamps:" {...}}
}

C Supplementary Results

C.1 Masking Control

We conduct masking control experiments to investigate whether the generated pseudo-queries rely excessively on specific attributes of historical target items. Specifically, we mask the corresponding attributes during pseudo-query generation and compare AdaM-Rec with several competitive baselines on the Amazon Beauty subset. Rare attributes are defined as visual or textual attributes occurring in fewer than 1% of the items, such as `hypoallergenic`, `exposed_ev_led_array`, `chrome_like`, `tattoo_coverage`, and `prosthetic_use`. As shown in Table 5, AdaM-Rec maintains relatively stable performance across different masking settings and consistently outperforms the competitive baselines. These results suggest that the generated pseudo-queries exhibit good generalizability and do not excessively overfit to idiosyncratic attributes of historical target items in a manner that substantially degrades downstream recommendation performance. Furthermore, the **Overfitting** metric in our pseudo-query quality evaluation provides additional evidence that the generated pseudo-queries do not rely excessively on specific long-tail attributes, further supporting their generalizability.

Table 5: **Performance comparison under different masking controls on the Amazon Beauty subset.** Rare-Attribute Mask, Title Mask, and Brand Mask denote masking the corresponding attributes during pseudo-query generation. Bold values indicate the best results.

Method	HR ↑		NDCG ↑	
	@20	@40	@20	@40
MACF	0.1551	0.1872	0.0725	0.0791
TAIRA	0.1497	0.1925	0.0798	0.0886
AdaM-Rec (Full)	0.1711	0.2193	**0.0979**	**0.1077**
+ Rare-Attribute Mask	0.1711	0.2166	0.0975	0.1069
+ Title Mask	0.1658	0.2112	0.0938	0.1031
+ Brand Mask	**0.1738**	**0.2219**	0.0963	0.1062

C.2 Cold-Start Analysis

We conduct a cold-start analysis under conditions consistent with the standard ablation setting used in the manuscript to investigate the effectiveness of self-optimization with limited user interaction histories. Specifically, an N-shot cold-start setting retains only the N most recent interactions from each user's complete history to simulate limited historical context. We introduce two metrics to characterize the behavior of the modality router. **Preference Strength** is computed as the average absolute deviation of the routing weights (ranging from 0 to 1) from the neutral point of 0.5. Its value therefore ranges from 0 to 0.5, with higher values indicating stronger and more decisive modality preferences. **Modality Selection Accuracy** measures the proportion of cases in which the router assigns a higher weight to the modality that yields better recommendation performance. As shown in Table 6, under extreme cold-start conditions (e.g., the 1-shot setting), self-optimization may introduce bias due to insufficient historical information, resulting in performance degradation compared with the variant without self-optimization. However, as more user interactions become available, both Preference Strength and Modality Selection Accuracy consistently increase, accompanied by improvements in recommendation performance. Notably, with three or more interactions, self-optimization consistently outperforms the variant without self-optimization across all recommendation metrics. These findings suggest that although richer user histories remain beneficial, the effectiveness of self-optimization does not heavily depend on extensive historical interactions, demonstrating its ability to provide meaningful performance gains even with relatively limited user histories.

Table 6: **Cold-start analysis of self-optimization under varying amounts of user interaction history.** Preference Strength measures the decisiveness of modality preferences, while Modality Selection Accuracy evaluates the correctness of modality selection. Bold values indicate the best results for modality selection accuracy and recommendation performance.

Setting	Preference Strength	Modality Selection Accuracy ↑	Recall ↑	HR ↑		NDCG ↑	
				@20	@40	@20	@40
w/o Self-Optimization	–	–	0.3743	0.1524	0.1898	0.0787	0.0866
1-shot Cold Start	0.1228	0.6818	0.3556	0.1444	0.1764	0.0701	0.0796
2-shot Cold Start	0.2137	0.8075	0.3770	0.1497	0.1872	0.0758	0.0864
3-shot Cold Start	0.2675	0.8850	0.4011	0.1578	0.2005	0.0839	0.0949
4-shot Cold Start	0.2931	0.9011	0.4118	0.1631	0.2086	0.0896	0.1002
Full (Non-Cold Start)	0.3416	**0.9385**	**0.4278**	**0.1711**	**0.2193**	**0.0979**	**0.1077**

C.3 Embeddings for Candidate Recall

Table 7: **Performance comparison between structured profile embeddings and raw embeddings during the candidate recall stage.** All experiments follow the standard ablation setting. Bold values indicate the best results.

Recall Setting	Recall ↑	HR ↑		NDCG ↑	
		@20	@40	@20	@40
Structured Profile Embeddings	0.4118	0.1684	0.2112	0.0952	0.1037
Raw Embeddings	**0.4278**	**0.1711**	**0.2193**	**0.0979**	**0.1077**

To investigate the advantages of using raw embeddings rather than structured profile embeddings during the recall stage, we conduct a comparative experiment under conditions consistent with the standard ablation setting used in the manuscript. Specifically, we compare the recommendation performance of the two representations when used for candidate recall. As shown in Table 7, using raw embeddings during the recall stage consistently yields better performance across all evaluation metrics. We hypothesize that this advantage arises from the distinct properties of the two representations. Specifically, structured profiles distill the original textual and visual content of each item into cleaner and more compact summaries while preserving key semantic information. Such representations are particularly suitable for the subsequent LLM-based ranking stage, as they provide concise textual evidence and reduce interference from noisy multimodal content during preference-level matching. However, this abstraction process inevitably discards some fine-grained information. In contrast, raw embeddings are directly derived from the original item content and can therefore preserve richer semantic and fine-grained signals. Moreover, similarity-based matching in the embedding space can be relatively robust to noisy information, partly because strong embedding models are typically

trained on large-scale and naturally noisy data. Consequently, these less lossy representations are better suited for identifying potentially relevant candidates from a large item pool.

Nevertheless, the relatively small performance gap between the two representations suggests that the fine-grained information potentially lost during structured profile construction has a limited impact on recall performance. This further indicates that structured profiles can effectively capture most of the task-relevant information from the original multimodal content. Overall, these findings support our design choice of employing raw embeddings for candidate recall while leveraging structured profiles for the subsequent LLM-based ranking stage, allowing the two representations to complement each other across different stages of the recommendation pipeline.

C.4 Stability Analysis

We supplement the variances of the main results in Table 8 and 9 to assess the stability of the reported gains.

Table 8: **Variance ($\times 10^{-6}$) of the results** reported in Table 1.

Method	Amazon Beauty				Amazon Clothing				Amazon Music			
	HR		NDCG		HR		NDCG		HR		NDCG	
	@20	@40	@20	@40	@20	@40	@20	@40	@20	@40	@20	@40
ReAct	75.282	133.581	27.838	36.909	51.173	82.759	16.990	21.332	34.658	156.313	19.912	31.116
Reflexion	56.792	53.955	25.689	26.757	99.824	42.415	20.179	23.750	45.021	49.568	24.809	25.444
InteRecAgent	16.820	49.725	9.578	13.094	20.026	45.291	11.043	14.922	53.134	40.394	32.645	33.560
MACRS	26.570	30.335	9.012	10.049	13.794	29.948	7.494	8.738	25.935	17.671	7.415	8.119
MACRec	44.958	78.235	28.629	33.627	18.859	70.201	8.598	12.793	26.116	34.050	20.088	23.217
MACF	56.282	221.563	29.179	48.364	20.943	53.282	12.461	16.897	44.405	127.500	36.214	42.560
TAIRA	43.162	119.067	16.191	22.519	18.868	69.505	13.095	19.664	48.455	86.525	40.954	44.131
AdaM-Rec	37.011	90.875	10.082	15.510	69.170	19.965	14.103	21.941	94.908	61.097	30.834	32.807

Table 9: **Variance ($\times 10^{-6}$) of the results** reported in Table 2.

Method	Amazon Video Games				Amazon Baby Products			
	HR		NDCG		HR		NDCG	
	@20	@40	@20	@40	@20	@40	@20	@40
GRU4Rec	9.303	13.286	4.357	5.861	1.714	4.011	0.944	2.131
SASRec	3.978	5.744	1.372	1.815	2.004	3.945	0.567	0.869
MGAT	5.604	11.124	3.166	4.407	1.466	3.181	0.989	1.764
MMGCN	4.572	11.355	3.294	4.833	3.118	7.733	1.832	3.261
MMSR	7.698	14.437	4.772	6.567	2.985	6.895	1.666	2.905
Trans2D	3.858	8.045	1.347	2.477	2.229	1.604	0.649	0.719
MLLM-MSR	8.344	17.956	4.270	6.312	1.646	4.148	1.066	2.246
AB-Rec	3.887	2.844	1.130	1.156	1.241	2.041	0.581	0.604
LLaRa	6.570	7.959	3.020	3.363	2.554	6.373	0.855	1.379
LLMRec	2.217	2.595	1.326	1.569	1.376	1.394	0.495	0.636
TallRec	10.210	15.385	3.576	4.748	5.663	4.300	1.671	1.849
SPRec	6.030	11.728	3.199	4.394	4.819	5.307	1.932	2.293
RecZero	4.994	12.135	2.851	5.548	4.212	7.361	3.720	4.059
MLLMRec-R1	3.588	7.252	1.722	2.084	1.583	1.657	0.588	0.726
AdaM-Rec	4.545	5.779	2.580	3.157	3.163	2.651	1.394	1.591

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